Run report: item1D-mult-run-20261018T2106Z

Descriptive record of one recorded run of examples/decomposition_coverage.rs, mode mult (Amendment D) under `research/weil_descent_angles_20261018/PROTOCOL.md` (Addendum A and Amendments B–G, each committed before the run it governs). It lists what ran, on what host, and what the tool recorded, file by file. Interpretation is in the study's `REPORT.md`. Costs are counted operations (F_p multiplications and inversions, group operations, native operations), never wall-clock; the only times are the tool's own seconds. Isolation: none earned (a shared cloud container, 4 threads); this affects only the seconds, since every recorded value is an exact count or a counted operation total.

env.txt

```
commit 44a81248a6aaf64ac4d5373e0c97867675887e2f
cmd: dc research/weil_descent_angles_20261018/results/item1D-mult-run-20261018T2106Z mult
rustc 1.97.0 (2d8144b78 2026-07-07)
Sun Oct 18 21:06:02 UTC 2026
post-run (d2edb2ab8): the tool was rustfmt-formatted (format only; unit tests pass); outputs unaffected.
```

checks.txt

```
M1 identity x(P+Q)+x(P-Q) = x1x2/(x1+x2)^2 violations: 0
M2 solution cosets not a power of 2: 0
M3 cross-check distinct(subfield) != distinct from cosets: 0 curves
M3 dispersion char 0 null: mean 0.9923 sd 0.0213 over 15 curves
M3 dispersion char 0 subfield: mean 0.9725 sd 0.0320 over 15 curves
M3 dispersion char 2 null: mean 0.9576 sd 0.0955 over 21 curves
M3 dispersion char 2 subfield: mean 1.0007 sd 0.0301 over 21 curves
```

binary_cosets.csv (first 12 of 21 rows)

m	curve	order	m1_pairs	m1_violations	x3_total	solvable	nonpow2_cosets	kdim0	kdim1	kdim2	kdim3	kdim4plus	valid0	valid1	valid2	valid3	valid4	valid5	valid6	valid7	valid8plus	distinct_subfield	distinct_from_cosets
4	0	240	28	0	256	120	0	0	240	0	0	16	79	26	15	0	0	0	0	0	0	83	83
4	1	234	36	0	256	120	0	0	240	0	0	16	70	28	22	0	0	0	0	0	0	101	101
4	2	240	28	0	256	120	0	0	240	0	0	16	79	26	15	0	0	0	0	0	0	83	83
5	0	990	136	0	1024	496	0	0	992	0	0	32	291	138	67	0	0	0	0	0	0	411	411
5	1	990	136	0	1024	496	0	0	992	0	0	32	300	120	76	0	0	0	0	0	0	393	393
5	2	1034	136	0	1024	496	0	0	992	0	0	32	286	148	62	0	0	0	0	0	0	421	421
6	0	3982	528	0	4096	2016	0	0	4032	0	0	64	1235	506	275	0	0	0	0	0	0	1563	1563
6	1	4034	528	0	4096	2016	0	0	4032	0	0	64	1245	486	285	0	0	0	0	0	0	1543	1543
6	2	4096	496	0	4096	2016	0	0	4032	0	0	64	1269	502	245	0	0	0	0	0	0	1495	1495
7	0	16492	2016	0	16384	8128	0	0	16256	0	0	128	5075	2074	979	0	0	0	0	0	0	6107	6107
7	1	16230	2080	0	16384	8128	0	0	16256	0	0	128	5016	2064	1048	0	0	0	0	0	0	6225	6225
7	2	16408	2016	0	16384	8128	0	0	16256	0	0	128	5103	2018	1007	0	0	0	0	0	0	6051	6051

mult.csv (first 12 of 72 rows)

char	q	curve	order	arm	F	lambda	poisson	pi	dispersion	h0	h1	h2	h3	h4	h5	h6	h7	h8plus
2	16	0	240	subfield	15	0.500000	0.393469	0.345833	1.06710	157	52	30	0	0	0	0	0	0
2	16	0	240	null	10	0.229167	0.204804	0.212500	0.79079	189	50	0	0	0	0	0	0	0
2	16	1	234	subfield	17	0.653846	0.479958	0.431624	0.99309	133	56	44	0	0	0	0	0	0
2	16	1	234	null	20	0.897436	0.592387	0.636752	0.82163	85	110	26	10	2	0	0	0	0
2	16	2	240	subfield	15	0.500000	0.393469	0.345833	1.06710	157	52	30	0	0	0	0	0	0
2	16	2	240	null	13	0.379167	0.315568	0.333333	0.76758	160	74	5	0	0	0	0	0	0
2	32	0	990	subfield	33	0.566667	0.432586	0.415152	0.94260	579	276	134	0	0	0	0	0	0
2	32	0	990	null	36	0.672727	0.489685	0.510101	0.83862	485	376	112	16	0	0	0	0	0

2	32	1	990	subfield	33	0.566667	0.432586	0.396970	1.00877	597	240	152	0	0	0	0	0	0
2	32	1	990	null	26	0.354545	0.298508	0.308081	0.87126	685	272	30	2	0	0	0	0	0
2	32	2	1034	subfield	33	0.542553	0.418738	0.407157	0.92926	613	296	124	0	0	0	0	0	0
2	32	2	1034	null	22	0.244681	0.217046	0.192456	1.16243	835	158	36	4	0	0	0	0	0

Graph

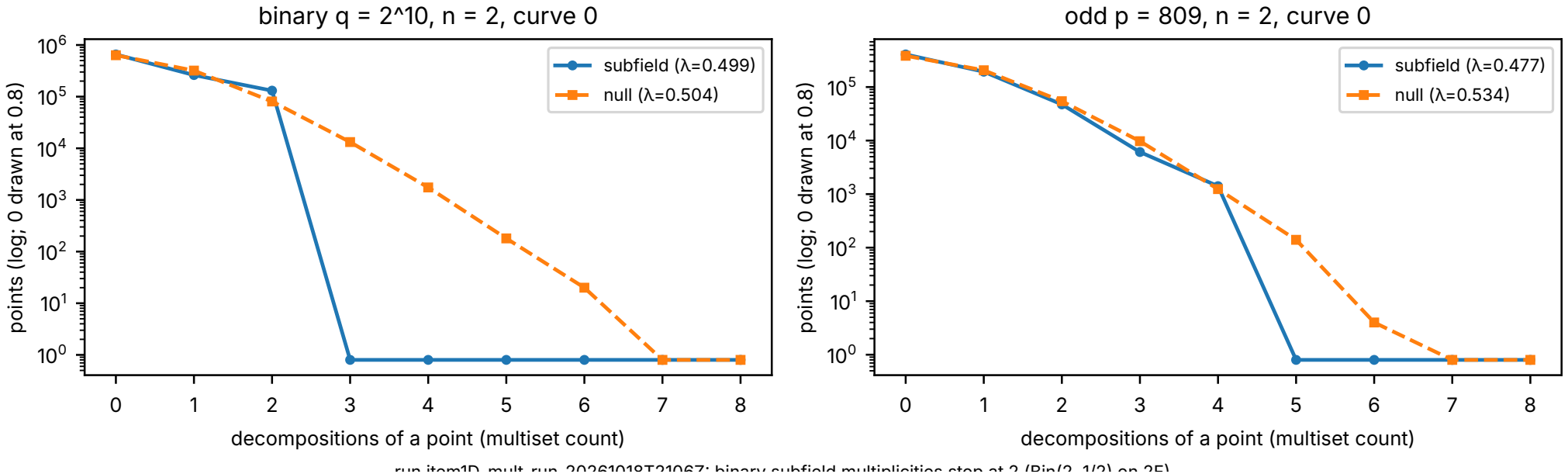


Figure 1: Amendment D: point multiplicity histograms, subfield vs null; exact.